Optimization Basics, Chebyshev Approximation and Conic Fitting

Convex Optimization, Homework 1

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English summary of my original Persian report, which is kept as a PDF in this folder.

Conceptual questions

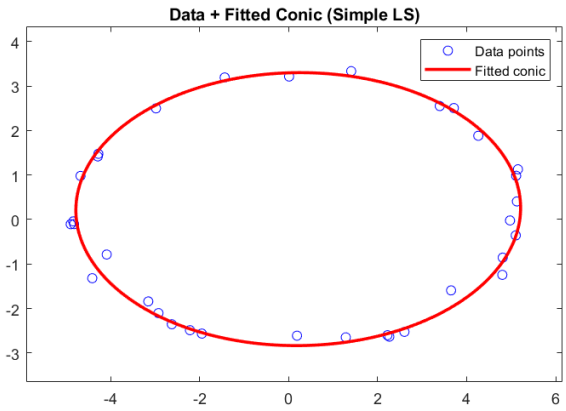
- **The standard problem.** In $\min f_0(x)$ s.t. $f_i(x) \le 0$, the decision variable lives in $\mathbb{R}^n$, the domains of the f_i are subsets of $\mathbb{R}^n$, and the functions are real-valued.
- **Chebyshev (minimax) approximation.** $\min_x \max_i |a_i^T x - b_i|$ minimises the worst error instead of the sum of squares. With an auxiliary variable t and the pairs of constraints $-t \le a_i^T x - b_i \le t$ it becomes an LP. It shows up in data fitting, filter design and robust control. I also sketched a minimax model for crowding in a metro station.
- **Infimum versus minimum.** $1/x$ on $x > 0$ has infimum 0 but no minimum, and it has no maximum.
- **Facility location.** I formulated the best location of a petrol station among three points as a location (minimax or median) problem. Under an equal-distance criterion the answer is the incentre of the triangle, which is a convex combination of the three points.
- **Why convexity matters.** Every local optimum is global, interior-point methods solve convex problems in polynomial time, duality gives sensitivity information, and many practical problems (least squares, LP, SDP) are convex or can be made convex. I also defined global optima, local optima and sub-optimal solutions.

Fitting a conic to data (`Q2.m`)

The points in `data.mat` should lie on a conic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. Each point gives one row $[x_i^2,\ x_i y_i,\ y_i^2,\ x_i,\ y_i,\ 1]$ of a matrix M. To avoid the trivial solution I fixed $F = 1$. That leaves an ordinary least-squares problem $\min_\alpha \|M'\alpha + \mathbf{1}\|_2$ for $\alpha = (A, B, C, D, E)$, where M' is the first five columns of M. Re-running the script gives

$$A = -0.0404, \quad B = 0.0011, \quad C = -0.1071, \quad D = 0.0173, \quad E = 0.0498, \quad F = 1,$$

an ellipse that passes through the data:



Lamp illumination

For the lamp problem I wrote three formulations:

- making one point brighter than the average illumination;
- minimising total lamp power when any 5 of the 10 lamps together may use at most a given fraction of the power;
- using a single lamp to light every point.